

MARCUS CHITURA

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Projects Portfolio - <https://marcuschitura.github.io/Projects-Portfolio/>

PROFESSIONAL SUMMARY

A graduate Electrical Engineer seeking a position as an entry level engineering professional, with a broad expertise across power systems, control systems and communication systems. Experienced in circuit and lighting design, embedded systems and computational modelling for real world engineering solutions. Equipped with system modelling skills in MATLAB/Simulink, FEMM, AutoCAD, Dialux and C++, with a proven ability to translate theoretical models into working designs through an array of multi-disciplinary projects implemented throughout the completion of my degree. Provided is an online portfolio showcasing selected engineering and research projects highlighting problem solving, system modeling, technical reporting and implementation skills.

WORK EXPERIENCE

March 2024 – April 2024

Research Assistant | University of Witwatersrand | Johannesburg, ZA

As part of an ongoing research initiative on magnetic circuit design, I was tasked with improving the accuracy of finite element models used in analysis. Using FEMM, I developed and validated magnetic circuit models, refining parameters to simulate realistic magnetic behavior. I conducted comparative analysis of multiple circuit configurations to evaluate efficiency and performance, then compiled a detailed simulation report summarising key findings and outcome.

November 2023 – February 2024

Engineering Intern | CGM Consulting Engineers | Harare, ZW

I contributed to several building services projects focusing on electrical system design, lighting layout optimisation, and project documentation. Using Dialux and AutoCAD, I designed room-specific lighting layouts that met client illumination requirements and SANS regulatory requirements. I also reviewed and corrected inconsistencies in Bills of Quantities, streamlining the verification process. My role further involved assisting with onsite inspections under senior supervision to ensure the implemented systems matched design intent and safety standards.

EDUCATION

Bachelor of Science in Electrical Engineering | University of the Witwatersrand

- Relevant Coursework: Electrical Design I & II, Control Systems I & II, Signals & Systems I & II, Software Development I & II, Power Engineering, Data Structures & Algorithms

TECHNICAL SKILLS

- Programming: C++, Python, MATLAB, R, Arduino IDE, Git, ARM Assembly Language
- Simulation & Design: FEMM, MotorSolve, LTSpice, Proteus Circuit Simulator, Simulink & MATLAB
- Electrical & CAD Tools: AutoCAD, Dialux Lighting Design, Proteus PCB Design
- General: Microsoft Office Suite, Quantitative Analysis, Technical Reporting, Problem Solvin